ASSIGNMENT – 1 (CO1)

STUDENT DETAILS

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Assignment	Assignment 1 – CO1

INTRODUCTION

Mobile communication networks provide voice and data connectivity to users spread across a geographical area. A cellular network divides the service region into smaller coverage areas called cells. Each cell is served by a base station, and radio channels are reused in other cells that are sufficiently far away to keep co-channel interference under control. Frequency reuse allows an operator to serve a very large number of subscribers even though the licensed spectrum is limited.

This assignment considers an industrial-technology township of **15 km²** containing engineering blocks, student hostels, staff residences, a commercial strip and a main ring road. The operator has **351 duplex voice channels** after control-channel overhead. The expected subscriber population grows from 2,000 to 10,000, 25,000 and eventually 120,000 or more. Each subscriber generates an average busy-hour traffic of **0.03 Erlang**, and the target grade of service is a call-blocking probability not greater than **2%**.

The assignment follows twelve tasks: designing alternative frequency-planning strategies, identifying the governing parameter of each, deriving the capacity and interference functions, analysing best, average and worst-case behaviour, scaling and sensitivity analysis, the cell-splitting recurrence and its solution, implementation and testing, presentation of results, comparison of theory with simulation, a final recommendation, and a day-4 design challenge involving a large event.

Design inputs at a glance

Parameter	**Value**
Township area	15 km²
Total duplex channels (S)	351
Traffic per subscriber	0.03 Erlang (busy hour)
Blocking target	≤ 2% (Erlang-B)
Minimum SIR	12 dB
Parameter	**Value**
Subscriber levels	2,000, 10,000, 25,000, 120,000
Reference cell radius (coverage case)	R = 0.6 km

TASK 1 – DESIGN ALTERNATIVE FREQUENCY-PLANNING STRATEGIES

1.1 Strategy 1: Fixed Cluster Frequency Reuse (FCA)

In Fixed Channel Assignment (FCA), the service region is divided into repeating clusters of N hexagonal cells. Each cell in a cluster receives a different group of channels, and the same group is reused in other clusters at a sufficient distance. The cluster size is chosen to maintain the required SIR while still leaving a reasonable number of channels per cell.

For this township the candidate cluster size is **N = 13** (i = 3, j = 1). The 351 channels are divided into 13 groups, so each cell receives 351/13 = **27 channels**. Smaller clusters such as N = 12 were examined first, but their ideal SIR at the conservative path-loss exponent n = 2.5 is only 11.67 dB, which falls below the 12 dB requirement. N = 13 is therefore the smallest valid cluster size that meets the threshold (12.11 dB).

Access-channel sharing

In FDMA each call receives its own frequency channel or frequency pair. In TDMA several users share one carrier by transmitting in different time slots. In OFDMA users are assigned groups of orthogonal subcarriers called resource blocks. Control channels carry access requests, paging and signalling, whereas traffic channels carry the actual voice or data. The 351-channel pool is treated as the traffic-channel resource that remains after control overhead.

Assumptions and design principle

- Cells are approximated as regular hexagons and the frequency pattern repeats consistently.
- Neighbouring cells exchange no channels under the basic fixed-assignment method.
- Traffic demand is predictable enough for a fixed channel group to be assigned to each cell.
- Co-channel cells are separated by the planned reuse distance.
- The governing principle is **spatial reuse**: the same spectrum is reused in geographically separated cells.

Iterative procedure / pseudocode

Valid hexagonal cluster sizes follow $N = i^2 + ij + j^2$, where i and j are non-negative integers. The procedure checks candidate sizes, computes the reuse ratio and SIR, and keeps only candidates that satisfy the threshold.

```
INPUT: total_channels = 351, minimum_SIR = 12 dB
FOR i = 0 to limit:
  FOR j = 0 to limit:
    N = i*i + i*j + j*j
    IF N > 0:
      Q = sqrt(3*N)
      Estimate SIR = (3N)^(n/2) / 6
      IF SIR >= minimum_SIR:
        channels_per_cell = floor(351/N)
        Estimate Erlang-B capacity per cell
        Store N, Q, channels_per_cell, capacity
Select the smallest feasible N and assign channel groups
Repeat the cluster pattern across the hexagonal grid
OUTPUT: feasible cluster sizes and frequency plan
```

FCA is worth investigating because it gives a stable reference design, simple frequency coordination and predictable interference. Its limitation is that a lightly loaded cell cannot use channels that remain idle in a neighbouring cell.

1.2 Strategy 2: Sectorised Frequency Reuse

Sectorisation divides the coverage area of a base station into directional sectors. A common arrangement uses three sectors of 120° each; a denser one uses six sectors of 60°. Each sector is served by a directional antenna that concentrates energy towards its own region and reduces the signal received from, or radiated towards, unwanted directions.

A user in one sector therefore sees fewer strong co-channel interferers than in an omnidirectional arrangement. Sectorisation does not create new licensed spectrum; it improves spatial isolation and can raise the usable SIR, subject to antenna side lobes, front-to-back ratio, sector overlap and neighbour coordination.

Access channels, assumptions and principle

Control and traffic resources are planned per sector. FDMA/TDMA or OFDMA may be used depending on the system. The design assumes that azimuth, mechanical or electrical tilt, sector boundaries and neighbour coordination are configured correctly. The governing principle is **spatial filtering**: antenna directivity reduces the contribution of unwanted interferers.

Pseudocode

```
FOR each base-station site:
  Divide the coverage region into 3 or 6 sectors
  Assign a coordinated channel group to each sector
  Configure antenna azimuth, tilt and transmit power
FOR each test-user position:
  Identify the serving sector
  Estimate wanted power using the serving-antenna gain
  Estimate interfering powers from each antenna pattern
  Calculate SIR and check the 12 dB requirement
Adjust sector orientation, tilt and power where required
OUTPUT: sector plan, SIR estimates, capacity indicators
```

Sectorisation is worth investigating around the hostels, the commercial strip, the ring road and the event ground, where demand is concentrated. It can improve SIR without buying more spectrum, but it raises antenna, radio-unit, installation and optimisation costs and creates additional handoff boundaries.

1.3 Strategy 3: Dynamic Channel Assignment and Borrowing (DCA)

Dynamic Channel Assignment does not reserve every channel permanently for one cell. A central or coordinated controller picks an available channel when a call arrives. A busy cell may borrow a channel from a lightly loaded neighbour if the channel is free and the assignment respects reuse-distance, interference and handoff constraints. When the call ends, the channel returns to the shared pool.

The controller checks channel occupancy, predicted interference, neighbouring load and resources reserved for handoffs or emergency signalling. The governing principle is **trunking efficiency and adaptive allocation**.

Assumptions and pseudocode

- Cells exchange timely channel-occupancy and load information.
- Allocation decisions are fast enough to serve call arrivals.
- Neighbouring cells follow common interference and reuse constraints.
- Borrowed channels are released safely when the call ends.
- Some channels are reserved for handoff or control signalling.

```
WHEN a call request arrives in cell c:
  List channels free in c and in eligible neighbouring cells
```

```
  Remove channels that violate reuse or interference limits
  Remove channels reserved for handoff/control
  Select the eligible channel with the lowest predicted interference
  IF a channel is available: assign it and update occupancy records
  ELSE: queue the request if supported, otherwise block it
WHEN the call ends:
  Release the channel and update the channel pool
```

DCA is attractive because traffic is not uniform: hostels peak at night, engineering blocks during the day, and the event ground during functions. Dynamic allocation can reduce blocking caused by idle channels stranded in lightly loaded cells, but it cannot enlarge the licensed spectrum and it must preserve SIR.

1.4 Preprocessing and cost responsibility

Before choosing a strategy the operator must prepare a land-use map, building-height map, traffic-density map, candidate-site list, road and access map, hexagonal grid and a radio-propagation survey. Busy-hour traffic must be estimated separately for engineering blocks, hostels, residences, roads and the event ground. Drive or walk tests are needed to find coverage holes, shadowing and unexpected interference.

The licensed operator normally bears the cost of traffic surveys, propagation measurements, radio planning, engineering design, site acquisition, equipment, installation and optimisation. The township authority may help by giving access to rooftops, poles, campus roads and land, but the operator stays responsible for network performance, spectrum compliance and service quality.

Summary of the three strategies

Strategy	Principle	Main advantage	Main limitation
Fixed cluster reuse	Spatial reuse	Simple, predictable channel plan	Idle channels may be stranded
Sectorisation	Spatial filtering	Reduces dominant interference	More equipment and sector coordination
Dynamic assignment	Trunking and adaptive allocation	Uses spare channels in nearby cells	Needs fast coordination and interference checks

TASK 2 – IDENTIFY THE GOVERNING DESIGN PARAMETER

Every frequency-planning strategy is controlled by one parameter that strongly influences capacity, interference or resource utilisation. The analysis concentrates on these parameters because they decide whether the design can meet the traffic and SIR targets.

Strategy	Governing parameter	Why it governs	Response to growth
Fixed cluster reuse	Cluster size N and reuse ratio Q = D/R	N sets the reuse distance, the SIR and the channels per cell	Choose the smallest feasible N; add cells or reduce radius as demand rises
Sectorisation	Number of sectors and effective interferer count	More directional sectors can cut strong interferers but add boundaries and handoffs	Increase sectorisation only in hotspots where antenna isolation is practical

Dynamic assignment	Available channel pool and per-cell offered traffic A	The controller can use eligible idle channels from other cells	Improve load reporting, borrowing rules and interference coordination

2.1 Fixed cluster reuse

For FCA, N fixes the number of channel groups. With S = 351 channels the nominal channels per cell are S/N. Increasing N raises the reuse distance and generally improves SIR, but cuts the channels per cell. The reuse ratio is Q = √(3N), which for N = 13 equals 6.245. The chosen value must satisfy the SIR threshold without making per-cell capacity too small.

2.2 Sectorisation

For sectorisation the effective number and strength of co-channel interferers matter most. A directional antenna suppresses signals arriving from outside its main beam. The effective interference depends on the radiation pattern, side lobes, sector orientation, tilt and user location. More sectors may improve isolation, but they also add sector boundaries and make handoff planning harder.

2.3 Dynamic channel assignment

For DCA the key quantity is the offered traffic A relative to the number of eligible channels. A cell with high offered traffic and few free channels has a high blocking risk. A lightly loaded neighbour may have spare resources, but they can only be borrowed if the reuse and SIR checks are satisfied. As the subscriber population grows, the controller needs more accurate load information and more frequent coordination.

Township-wide offered traffic is 60 Erlangs for 2,000 subscribers and 300, 750 and 3,600 Erlangs at 10,000, 25,000 and 120,000 subscribers. The network must therefore evolve from a coverage-led macro layout to a capacity-led hierarchical layout.

TASK 3 – DERIVE CAPACITY AND INTERFERENCE FUNCTIONS MATHEMATICALLY

3.1 Hexagonal geometry and cluster size

A regular hexagonal cell has six neighbours. A cluster contains N cells and is defined by two integer shift parameters i and j: move i cells along one lattice direction and j cells along a direction 60° away. This gives

$$N = i^2 + ij + j^2$$

Examples: N = 3 for (1,1), N = 4 for (2,0), N = 7 for (2,1), N = 9 for (3,0), N = 12 for (2,2) and N = 13 for (3,1). All are valid cluster sizes on a hexagonal grid.

3.2 Reuse distance and reuse ratio

Let R be the cell radius from centre to vertex. The centre-to-centre distance between co-channel cells is

$$D = R\sqrt{(3N)} \text{ and } Q = D/R = \sqrt{(3N)}$$

For N = 13: Q = √39 = 6.245, so D = 6.245R. A larger cluster pushes co-channel cells farther apart and generally lowers co-channel interference.

3.3 Hexagonal cell area and number of cells

A regular hexagon splits into six equilateral triangles of side R, each of area $(\sqrt{3}/4)R^2$.
Therefore

$$A_cell = 6 \times (\sqrt{3}/4)R^2 = (3\sqrt{3}/2)R^2 \approx 2.598\ R^2$$

The approximate number of cells to cover area A is C = ceil(A / A_cell). For R = 0.6 km, A_cell = 2.598 × 0.6² = 0.9353 km², so for 15 km² we get C = ceil(15/0.9353) = **17 cells**. This is a geometric coverage estimate only; real planning adds overlap, edge coverage, reserve capacity and extra cells in hotspots.

3.4 Channels per cell

If S = 351 channels are shared equally by the N cells of a cluster, the nominal channel count per cell is m = S/N. For N = 13, m = 351/13 = 27 channels. Where S/N is not an integer, whole channels must be allotted and the total must not exceed 351.

Cluster size N	**Channels per cell (⌊351/N⌋)**	**Reuse ratio Q = √(3N)**	**Ideal SIR at n = 2.5 (dB)**
3	117	3.000	4.15
4	87	3.464	5.71
7	50	4.583	8.75
9	39	5.196	10.11
12	29	6.000	11.67
13	27	6.245	12.11

3.5 Erlang-B traffic capacity

Erlang-B models a loss system in which an arriving call is blocked when every traffic channel is busy. It assumes no waiting queue and uses the offered traffic A and the number of channels m. The recursion is

$$B(0, A) = 1$$

$$B(m, A) = A \cdot B(m-1, A) \,/\, [\, m + A \cdot B(m-1, A)\,]$$

For m = 27 channels and a blocking target of 0.02, repeated evaluation (bisection on A) gives a maximum offered load of about **19.265 Erlangs per cell**. This is the planning limit used in the remaining calculations.

3.6 Township traffic and required cells

Busy-hour offered traffic is the number of subscribers multiplied by the traffic per subscriber: A_total = subscribers × 0.03 Erlang.

Subscribers	**Total offered traffic**	**Cells at 19.265 E/cell**	**Interpretation**
2,000	60 E	4	Coverage and minimum spacing dominate
Subscribers	**Total offered traffic**	**Cells at 19.265 E/cell**	**Interpretation**

10,000	300 E	16	Capacity begins to require extra cells
25,000	750 E	39	More capacity sites are needed
120,000	3,600 E	187	Dense small-cell layer is required

For 120,000 subscribers, A_total = 120,000 × 0.03 = 3,600 Erlangs. Required cells = ceil(3,600/19.265) = **187**. If 187 equal hexagons cover 15 km², the radius is

$$R \approx \sqrt{[\ 15 / (187 \times 2.598)\ ]} \approx 0.176\ km$$

This is a first-order estimate that excludes overlap, uneven demand, reserve capacity and real site constraints.

3.7 Co-channel SIR derivation for fixed reuse

Assume a mobile at distance R from its serving base station and six first-tier co-channel base stations each about D away. Received power varies as distance raised to the negative path-loss exponent n: the wanted power $\propto R^{(-n)}$ and each interferer $\propto D^{(-n)}$, so total interference $\propto 6D^{(-n)}$.

$$SIR \approx R^{(-n)} / [6\ D^{(-n)}] = (D/R)^{n} / 6$$

Since D/R = √(3N), this becomes

$$SIR \approx (3N)^{(n/2)} / 6\ \ SIR_{dB} = 10\ log_{10}(SIR)$$

For N = 13, 3N = 39. At n = 2.5, SIR = 39^1.25/6 = 16.24, i.e. **12.11 dB**. At n = 3, SIR = 40.59 (16.08 dB). At n = 4, SIR = $39^2/6$ = 253.5 (24.04 dB). The n = 2.5 open-campus case is the conservative one and leaves only about 0.11 dB of margin over the 12 dB threshold.

3.8 SIR under sectorisation

Sectorisation changes the interference denominator because not every co-channel base station contributes equally to a directional user. A general expression is

$$SIR_{sector} = P_{wanted} / \Sigma\ (\ P_{interferer,k} \times G_k\ )$$

Here G_k is the effective antenna gain towards interferer k relative to the wanted direction. A sector antenna reduces the weighted interference sum and so improves SIR. As an illustration, if sectorisation leaves an effective two interferers instead of six, SIR = (3N)^(n/2)/2, which for N = 13 and n = 2.5 gives 16.88 dB. It is not valid to assume that three-sector operation always divides the interferer count by exactly three: real patterns, side lobes, geometry and boundaries must be measured.

3.9 SIR under dynamic channel assignment

Dynamic assignment does not change the physical propagation equation. The controller only selects a channel when the candidate satisfies the reuse and interference constraints, so physical SIR is still set by wanted and interfering received powers. The benefit of DCA is better use of idle channels and possibly lower blocking, not extra spectrum.

3.10 Cost–capacity relationship

Let C_survey be the one-time cost of traffic surveys and propagation measurements, C_site the cost of one installed site, C_sector the extra sector-equipment cost per site and C_control the controller or software cost. For M sites a simplified model is

$$C_total = C_survey + M \cdot C_site + M \cdot C_sector + C_control$$

If each cell supports A_B Erlangs, aggregate capacity is about M·A_B, so

$$Cost\ per\ supported\ Erlang = C_total / (M \cdot A_B)$$

FCA has low control complexity but may need more sites when traffic is uneven. Sectorisation adds antenna and radio cost but can improve SIR and capacity per site. DCA adds controller and coordination cost but can reduce unused capacity in lightly loaded cells. Exact currency values cannot be calculated because equipment prices, rent, power and survey rates are not supplied.

TASK 4 – ANALYSE BEST, AVERAGE AND WORST-CASE BEHAVIOUR

Radio performance varies with the mobile's position inside the cell and with the traffic load at the time. A user close to the base station receives a stronger wanted signal than one at the cell edge, while the number and strength of visible co-channel interferers depend on geometry, antenna patterns and propagation.

Case	**Mobile position and traffic**	**SIR behaviour**	**Blocking behaviour**
Best	Near the serving base station; off-peak	Short wanted-signal distance gives high received power and generally high SIR	Few channels occupied; blocking close to zero
Average	Typical locations; average busy hour	Wanted and interfering powers are moderate	Blocking approaches the design target only near the planned load
Worst	Cell edge or corner; event surge	Wanted signal is weak and several neighbours may interfere	Channels fill quickly and blocking rises

4.1 Best-case analysis

Near the base station the wanted-signal distance d is less than the nominal edge radius R. Under the path-loss model, wanted power ∝ d^(−n), so a smaller d gives a stronger signal; if the interference geometry is unchanged, SIR rises. Off-peak, few channels are occupied, so the chance that a new call finds every channel busy is very low.

4.2 Average-case analysis

At ordinary positions the wanted signal is neither strongest nor weakest, and SIR depends on distance, the directions of co-channel cells and local shadowing. During the average busy hour occupancy rises. The Erlang-B blocking probability depends on both A and m, and it climbs towards 2% as the offered traffic approaches the 19.265 Erlang planning limit for 27 channels.

4.3 Worst-case analysis

At a cell edge or corner the mobile is far from its serving base station and may be close to more than one neighbour. Wanted power falls and interference grows. If an event surge also lifts offered traffic, all channels may be busy much more often, creating both a radio-quality problem and a capacity problem.

4.4 Numerical illustration and shadowing

For N = 13 and n = 2.5 the six-equal-interferer model gives SIR ≈ 12.11 dB, only 0.11 dB above the 12 dB target. If shadowing lowers the wanted signal by 6 dB while interference is unchanged, the effective SIR drops to about 6.11 dB, well below target. A design with such a thin margin must therefore be validated by field measurements and antenna optimisation.

A higher path-loss exponent reduces distant co-channel interference faster and raises the ideal SIR. In a dense built-up area, however, a high exponent also signals stronger attenuation and penetration loss, so buildings may create shadowed areas and coverage holes. A higher calculated SIR therefore does not by itself guarantee a better user experience.

The township-wide totals are 60, 300, 750 and 3,600 Erlangs. Blocking cannot be inferred from these totals alone, because it depends on how traffic is distributed among cells and on the channels available in each. An event ground can be overloaded even when the township average looks acceptable.

TASK 5 – SCALING AND SENSITIVITY ANALYSIS

Scaling analysis describes how capacity, interference and site count change when the cluster size, cell radius, subscriber population or propagation conditions change. It matters because the network must serve a growing population without receiving additional spectrum.

Change	**Mathematical effect**	**Planning interpretation**
N increases	Channels/cell ∝ 1/N; ideal SIR ∝ N^(n/2)	Better interference protection but fewer channels per cell
R decreases	Cell count ∝ $1/R^2$	More sites and higher aggregate capacity, but more handoffs and cost
Subscribers increase	Offered traffic ∝ subscribers	Capacity must rise through smaller cells, sectorisation and/or DCA
n increases from 2.5 to 4	Ideal SIR increases for fixed N	Distant interference falls faster, but coverage behind obstacles may worsen

5.1 Effect of cluster size

For fixed spectrum S, channels per cell are S/N, so a larger N leaves fewer channels per cell. The reuse ratio Q = √(3N) grows, and the ideal SIR is proportional to N^(n/2). Moving from N = 12 to N = 13 at n = 2.5 lifts SIR from 11.67 dB to 12.11 dB (just over the threshold) but cuts the per-cell capacity from 21.04 to 19.26 Erlangs. The designer must balance interference protection against traffic capacity.

5.2 Effect of cell radius

Hexagon area is proportional to R^2. For a fixed township area the number of cells is therefore proportional to $1/R^2$. Reducing R increases the number of cells and allows more spatial reuse, but it also increases site acquisition, backhaul, power, maintenance and handoff activity.

5.3 Effect of subscriber growth

Offered traffic grows linearly with subscribers under the fixed 0.03 Erlang assumption. The 120,000-subscriber case needs about 187 cells at 19.265 Erlangs per cell, far above the 17-cell geometric coverage estimate for R = 0.6 km. Coverage alone is not enough; the design becomes capacity-driven.

5.4 Break-even point

A cell reaches its planned capacity when offered traffic nears 19.265 Erlangs for 27 channels at 2% blocking. At or beyond that point more capacity must come from cell splitting, sectorisation or channel borrowing. Beyond 120,000 subscribers a macro-only design is unlikely to be economical or adequate; a hierarchical macro–micro/small-cell arrangement keeps wide-area coverage while concentrating capacity in hotspots.

Cluster size N	**Channels/cell**	**Capacity (E/cell)**	**SIR n = 2.5**	**SIR n = 3**	**SIR n = 4**
7	50	40.26	8.75	12.05	18.66
9	39	30.08	10.11	13.69	20.85
12	29	21.04	11.67	15.56	23.34
13	27	19.26	12.11	16.08	24.04

TASK 6 – FORMULATE THE RECURRENCE FOR REPEATED CELL SPLITTING

Cell splitting replaces an overloaded cell by smaller cells. Here the radius is reduced from R to R/2. Since area $\propto R^2$, each child covers one quarter of the parent area, so one parent cell is replaced by four child cells.

Let C_k be the number of cells after k uniform splitting rounds, K_k the ideal aggregate traffic capacity, R_k the cell radius, P_k the transmit power per cell and H_k the relative handoff rate. The base case is the original network at k = 0.

Quantity	**Recurrence**	**Closed form**
Cell count	C_k = 4·C_(k−1)	C_k = C_0 · 4^k
Aggregate capacity	K_k = 4·K_(k−1)	K_k = K_0 · 4^k
Cell radius	R_k = R_(k−1) / 2	R_k = R_0 / 2^k
Transmit power	P_k = P_(k−1) / 2^n	P_k = P_0 / 2^(nk)
Relative handoff rate	H_k ≈ 2·H_(k−1)	H_k ≈ H_0 · 2^k

6.1 Explanation of the recurrence

The cell-count recurrence holds because every parent is replaced by four children. The same factor of four is used for ideal aggregate capacity, under the simplifying assumption that each new cell provides a similar traffic-capacity unit and that the traffic distribution stays suitable. The radius halves each round.

The power relation follows the planning model P′ = P/2^n: when the required radio distance is halved, the power needed to keep a comparable received level falls according to the path-loss exponent. The relative handoff rate is approximated as doubling because smaller cells create more boundaries and more frequent transitions.

6.2 Selective splitting

Uniform splitting is not always economical because only some areas are overloaded. Let f be the fraction of current cells chosen for splitting. Each chosen cell becomes four cells, adding three cells per selected parent, so

$$C_k = C_{(k-1)} (1 + 3f) \; K_k = K_{(k-1)} (1 + 3f)$$

If the fraction varies by round, C_k = C_0 × Π(1 + 3f_j). Selective splitting suits this township, where engineering blocks, hostels and the event ground need more capacity than residential areas.

6.3 Work and cost per round

Each round needs site planning, installation, antenna configuration, frequency coordination, backhaul provisioning, neighbour-list updates and field testing, and both work and cost grow with the number of cells. The design must also respect power limits, regulatory requirements, handoff performance and interference between the new small cells and the existing macro layer.

TASK 7 – SOLVE THE RECURRENCE AND INTERPRET THE RESULT

7.1 Uniform splitting by iteration

The uniform capacity recurrence is K_k = 4K_(k−1) with base case K_0. Substituting repeatedly gives K_1 = 4K_0, K_2 = 4(4K_0) = 4^2K_0 and K_3 = 4(4^2K_0) = 4^3K_0. After k rounds

$$K_k = K_0 \cdot 4^k$$

To support a forecast demand D we need K_0·4^k ≥ D, i.e. 4^k ≥ D/K_0. Taking logarithms,

$$k \geq \log_4(D / K_0) \; k_{min} = \text{ceil}(\log_4(D / K_0))$$

7.2 Numerical example

Assume one original cell supports K_0 = 19.265 Erlangs and the target demand is D = 250 Erlangs. Then k_min = ceil($\log_4$(250/19.265)) = ceil(1.85) = **2 rounds**.

After 2 rounds the ideal capacity is 19.265 × 4^2 = 19.265 × 16 = 308.236 Erlangs. The cell count is multiplied by 16, the radius becomes R_0/4, the transmit power per cell becomes P_0/4^n, and the relative handoff rate rises to about 4 times the original.

7.3 Selective splitting solution

For a constant split fraction f, the selective recurrence is K_k = K_0 (1 + 3f)^k. Requiring K_k ≥ D gives

$$k_min = ceil[\ log(D / K_0) / log(1 + 3f)\ ]$$

This applies for $0 < f \leq 1$ under the simplified scaling assumption; if f = 0 no capacity is added. For example, with f = 0.3 and the same D and K_0, log(250/19.265) / log(1.9) = 3.99, so **4 rounds** are needed, more rounds than uniform splitting but each round touches far fewer sites.

7.4 Interpretation

The mathematical minimum number of rounds is not automatically the economic optimum. Splitting increases the number of sites, backhaul links, handoff events, maintenance tasks and coordination effort. A practical design splits only overloaded areas, uses sectorisation where it gives real isolation, and uses DCA where neighbouring cells have spare eligible channels.

TASK 8 – IMPLEMENT AND TEST THE PLANNING STRATEGIES

A reproducible analytical simulator can be written in Python. It calculates valid cluster sizes, channels per cell, approximate SIR, Erlang-B blocking and the number of cells needed for the offered traffic. The program compares fixed reuse, a simplified sectorisation approximation and dynamic channel assignment. The sector and dynamic outputs are planning indicators, not full radio-network simulation results.

8.1 Python implementation

```
import math

TOTAL_CHANNELS = 351
BLOCKING_TARGET = 0.02
TRAFFIC_PER_USER = 0.03
MIN_SIR_DB = 12.0

def valid_cluster_sizes(limit=30):
    values = set()     for i
in range(limit):         for j
in range(limit):             n
= i*i + i*j + j*j
if 1 <= n <= limit:
values.add(n)     return
sorted(values)

def sir_db(N, path_loss_n, interferers=6):
    sir = (3*N) ** (path_loss_n/2) / interferers
return 10 * math.log10(sir)

def erlang_b(channels, offered):
    b = 1.0     for m in range(1,
channels + 1):         b = offered*b /
(m + offered*b)     return b

def max_erlangs(channels, target=BLOCKING_TARGET):
    low, high = 0.0, 1000.0
for _ in range(100):
        mid = (low + high) / 2         if
erlang_b(channels, mid) <= target:
            low = mid
else:
high = mid     return
low

def evaluate_fixed(N, n=2.5):
    channels = TOTAL_CHANNELS // N
    return channels, max_erlangs(channels), sir_db(N, n)

def evaluate_sectorised(N, n=2.5, effective_interferers=2):
```

```
    channels = TOTAL_CHANNELS // N
    return channels, max_erlangs(channels), sir_db(N, n,
effective_interferers)

def evaluate_dynamic(N, n=2.5):
    # Borrowing helps only when neighbouring channels pass interference
checks.     channels = TOTAL_CHANNELS // N
    return channels, max_erlangs(channels), sir_db(N, n)

def main():
    print("Valid cluster sizes:", valid_cluster_sizes())
for users in [2000, 10000, 25000, 120000]:
traffic = users * TRAFFIC_PER_USER
print("\nSubscribers:", users, "Traffic:", traffic)
for N in [3, 4, 7, 9, 12, 13]:
            m, cap, sir = evaluate_fixed(N, 2.5)
cells = math.ceil(traffic / cap)
print("N:", N, "channels/cell:", m,
                  "capacity/cell:", round(cap, 3),
                  "SIR dB:", round(sir, 2), "cells:", cells)
print("\nSectorised illustration:", evaluate_sectorised(13, 2.5, 2))
print("Dynamic assignment baseline:", evaluate_dynamic(13, 2.5))

if __name__ == "__main__":
main()
```

8.2 How to execute the program

- Install Python 3 on the computer.
- Open a code editor such as Visual Studio Code, or create a file named cellular_planning.py.
- Copy the complete program into the file and save it.
- Open a terminal in the same folder.
- Run the command: `python cellular_planning.py`
- Record the printed cluster sizes, channels per cell, Erlang capacity, SIR and estimated cell count.

8.3 Test matrix

Test dimension	**Values to test**
Cluster size N	3, 4, 7, 9, 12; N = 13 as the selected candidate
Path-loss exponent n	2.5, 3 and 4
Antenna layout	Unsectorised and sectorised
Subscriber population	2,000; 10,000; 25,000; 120,000
Strategies	Fixed reuse, sectorised reuse and dynamic assignment

The same user locations and random seed should be used when extending the program to a Monte-Carlo simulation. The full extension should generate random mobile positions in hexagonal cells, compute path loss and log-normal shadowing, apply sector antenna gains, select eligible channels and record the SIR of each trial. The compact program above is an analytical calculator and does not produce Monte-Carlo user-location results.

8.4 Output of the analytical calculator (computed)

N	**Channels/cell**	**Capacity (E/cell)**	**SIR n = 2.5 (dB)**

3	117	104.493	4.15
4	87	75.415	5.71
7	50	40.255	8.75
9	39	30.081	10.11
12	29	21.039	11.67
13	27	19.265	12.11

Cells required for N = 13 (19.265 E/cell): 4 cells at 2,000 subscribers, 16 at 10,000, 39 at 25,000 and 187 at 120,000. Sectorised illustration (two effective interferers): 16.88 dB.
Dynamic baseline: same physical SIR as fixed reuse, 12.11 dB.

8.5 Sanity check

For N = 7 and n = 4 the six-interferer approximation gives SIR = $(3\times7)^2/6 = 21^2/6 = 73.5$.
In decibels $10\log_{10}(73.5) \approx 18.66$ dB (about 18.7 dB), matching the program output.

TASK 9 – RECORD AND PRESENT EXPERIMENTAL PERFORMANCE

The simulator should record the same measurements for every strategy and traffic level so that radio-quality and capacity limits can be compared on a common basis. A result is not reproducible unless its input assumptions and simulation settings are stated.

Metric	**Meaning**	**Recommended presentation**
Mean SIR	Average signal-to-interference ratio	Mean value and SIR cumulative distribution function (CDF)
5th-percentile SIR	SIR exceeded by 95% of users	Table and CDF; indicates edge-user performance
Outage probability	Fraction of users below 12 dB SIR	Percentage bar chart
Blocking probability	Blocked call attempts ÷ total attempts	Blocking versus offered-traffic graph
Channels per cell	Traffic channels assigned to a cell	Comparison table
Sites required	Cells/sites needed for the offered load	Capacity versus subscriber graph
Runtime	Time to complete the simulation	Reproducibility table

9.1 SIR distribution

Mean SIR alone can hide poor performance near cell boundaries. The 5th-percentile SIR describes the weak-signal part of the user population. Outage probability is the percentage of users whose SIR is below 12 dB. A cumulative distribution function should be plotted to show the proportion of users at or below each SIR value.

9.2 Blocking probability

Blocking probability should be plotted against offered traffic for each strategy with the 2% target drawn as the design threshold. A strategy that keeps blocking lower under uneven traffic may be preferable even if its theoretical SIR is not the highest.

9.3 Reproducibility settings

- Random seed and number of user locations.
- Number of call attempts and the call-arrival model.
- Path-loss model and path-loss exponent.
- Shadowing standard deviation and penetration-loss assumptions.
- Antenna pattern, azimuth, tilt and sector configuration.
- Channel-allocation and borrowing rules.
- Cell layout, boundary conditions and simulation runtime.

The analytical values in this report are computed, but no Monte-Carlo outputs have been run. Numerical mean SIR, 5th-percentile SIR, outage percentage and runtime must therefore come from executing a fully specified simulation; they must not be invented or presented as measured results.

TASK 10 – COMPARE THEORY WITH SIMULATION

Analytical models give a simplified prediction of network behaviour, while simulation represents more realistic geometry, user positions and radio conditions. The two should show the same overall trends even when exact numbers differ.

10.1 Expected agreement

- A larger cluster size increases co-channel separation and improves the ideal SIR.
- A larger cluster size reduces the channels assigned to each cell.
- Smaller cells increase site count and raise aggregate capacity through spatial reuse.
- Sectorisation reduces dominant interference when antenna directivity is effective.
- Dynamic assignment reduces blocking when neighbouring cells have spare eligible channels.

10.2 Sources of deviation

- The six-interferer model assumes six equal-distance interferers, whereas real cells have different distances and powers.
- Shadowing and building penetration randomly change signal strength.
- Finite network edges have fewer interferers than an infinite repeating grid.
- Users are not uniform: lecture halls, hostels and the event ground form traffic hotspots.
- Erlang-B assumes independent arrivals and no waiting queue; real voice and data traffic may differ.
- Integer rounding means S/N is not always a whole number (for example 351/4 = 87.75).
- Real sector antennas have side lobes and imperfect front-to-back isolation.
- Mobility and handoff behaviour change the resources a user sees.

10.3 Strategy comparison

FCA is a stable baseline because its allocation and interference pattern are predictable. Sectorisation helps where directional traffic and antenna isolation can cut dominant interferers. DCA helps where neighbouring cells have uneven traffic and spare channels can be borrowed safely. A hybrid network can combine all three advantages.

The best strategy must be chosen using blocking, 5th-percentile SIR, outage, site count, runtime and total cost together, not simply because it has the highest theoretical capacity. Simulation results should be compared with the analytical predictions after the actual experiment has been run.

TASK 11 – RECOMMEND AND DEFEND THE FINAL DESIGN

11.1 Recommended network architecture

The recommended baseline is a hierarchical cellular network. Fixed frequency reuse gives the stable macro-layer plan. Sectorisation is added in high-demand zones, and dynamic channel assignment or borrowing is used where neighbouring cells have uneven traffic. Microcells or small cells are introduced as demand grows, particularly around hostels, engineering blocks, the commercial strip and the event ground.

11.2 Selected planning parameters

Parameter	**Recommended baseline**	**Reason**
Cluster size	N = 13	Smallest valid reuse pattern that meets the 12 dB SIR target at n = 2.5
Total channels	351	Fixed licensed allocation
Channels per cell	27	351/13
Blocking target	≤ 2%	Specified grade of service
Traffic capacity per cell	≈ 19.265 Erlangs	Erlang-B limit for 27 channels at 2% blocking
Minimum SIR	12 dB	Selected digital-style threshold
Path-loss planning case	n = 2.5	Conservative open-campus case; lowest of the three SIR estimates

11.3 Expected capacity and interference

For N = 13 the approximate SIR values are 12.11 dB at n = 2.5, 16.08 dB at n = 3 and 24.04 dB at n = 4. The open-campus case has only a 0.11 dB margin over the 12 dB threshold, so careful antenna optimisation, site placement and field validation are required. Adding sectorisation is the most direct way to widen this margin.

With 27 channels per cell the offered-traffic limit is about 19.265 Erlangs at 2% blocking. For 120,000 subscribers the total offered traffic is 3,600 Erlangs and the first-order estimate requires 187 cells, equal hexagons of radius about 0.176 km covering 15 km².

11.4 Cost and operational trade-offs

The macro layer provides broad coverage and avoids excessive site deployment in low-demand areas. Small cells add capacity in concentrated zones but increase site rent, power, backhaul and maintenance. Sectorisation improves spatial filtering but adds antenna and radio equipment. Dynamic borrowing improves trunking efficiency but needs reliable load reporting and interference coordination. The operator should compare cost per supported Erlang and the quality of service achieved.

11.5 Growth beyond 120,000 subscribers

A single macro-cell layer is not enough for the 120,000-subscriber planning case under the stated traffic assumptions. The network must keep densifying and optimising capacity. A hierarchical macro–micro/small-cell layout is more practical: macro sites hold wide-area coverage while smaller cells serve local hotspots. The design stays feasible only if the operator keeps adding capacity and coordinating interference as demand grows.

11.6 Final defence

The hybrid recommendation is stronger than relying on any single strategy. FCA gives a predictable baseline, sectorisation improves spatial isolation in busy zones, and DCA uses eligible idle channels where traffic is uneven. Cell splitting adds capacity when a cell nears its Erlang-B limit. Together they address interference, blocking, mobility and cost rather than maximising one metric.

TASK 12 – DAY-4 DESIGN CHALLENGE: RE-EVALUATE AND ADAPT THE DESIGN

12.A Impact analysis

The changed scenario introduces a convocation or sports event with 30,000 attendees in a 1.5 km² event zone. Each attendee generates 0.04 Erlang, the event lasts four hours, and such events occur about twice a month. The event traffic is

$$A_event = 30{,}000 \times 0.04 = 1{,}200 \text{ Erlangs}$$

The event traffic density is 1,200 / 1.5 = 800 Erlangs/km². For comparison, the 120,000-subscriber township-wide traffic is 3,600 Erlangs over 15 km², an average of 240 Erlangs/km². The event zone therefore carries 3.33 times the township-wide average density.

The assumptions of uniform traffic, steady demand and permanent capacity no longer hold. The surge is concentrated in a small area and lasts a limited time. Since the event is predictable and occurs only about twice a month, temporary capacity is likely to be more economical than permanently redesigning the whole township.

12.B Evaluation of the original design

The original N = 13 design gives each cell 27 channels and supports about 19.265 Erlangs per cell at 2% blocking. The cell-equivalents needed for the event are ceil(1,200/19.265) = **63**.

The original macro layout cannot safely carry this concentrated demand without extra capacity. Channel exhaustion and rising blocking will appear first in the event-ground cells, and high user density also raises interference, signalling load and handoff activity. Township-wide spare capacity cannot automatically fix the hotspot, because spare channels can be borrowed only when they are free and satisfy interference and reuse constraints. Dynamic borrowing may help but cannot be assumed to supply the full 1,200 Erlangs.

12.C Adaptation or redesign

Decision: MODIFY TEMPORARILY. Keep the normal township macro network and deploy temporary capacity around the event ground on event days, using temporary microcells or cells-on-wheels (COWs), sectorised antennas and coordinated dynamic channel borrowing.

- Pre-plan temporary frequency groups, backhaul, power supply, site permissions and handoff boundaries.
- Use about 63 cell-equivalents as a first-order minimum at 19.265 Erlangs per cell, with extra headroom for bursty demand.
- Divide the 1.5 km² zone into 63 equal hexagonal areas for an illustrative radius estimate.
- Use 60° or 120° sectorisation where directional traffic flows justify it.
- Adjust tilt and transmit power to limit overshoot and interference into the macro layer.
- Use dynamic borrowing only after checking co-channel interference and reserving resources for handoff and control.
- After the event, switch off or redeploy the temporary cells to avoid permanent operating cost.

For 63 equal hexagonal areas in 1.5 km², the illustrative radius is

$$R \approx \sqrt{[\ 1.5 / (63 \times 2.598)\ ]} \approx 0.096\ km \approx 96\ metres$$

This is a theoretical density estimate, not a final site map. Real deployment depends on stadium geometry, audience distribution, antenna mounting positions, safety rules and backhaul access.

12.D Evidence-based defence

Temporary modification is preferred because the event is predictable, confined to 1.5 km² and lasts only four hours, while event days occur about twice a month. Permanent event-ground densification would incur site rent, power, maintenance and backhaul costs even when the ground is empty. Temporary COWs and microcells put the spending where and when it is needed.

The traffic calculation supports this: 1,200 Erlangs is far beyond the capacity of a few macro cells and needs roughly 63 cell-equivalents at the selected Erlang-B limit. Sectorisation and dynamic borrowing can improve utilisation, but the operator must still supply enough radio resources and backhaul. Final acceptance should require measured SIR, outage and blocking during a controlled event trial.

12.E Risks and operational controls

Temporary cells must be integrated carefully with the permanent macro network. Poorly planned transmit power can cause overshoot and extra co-channel interference. Wrong neighbour lists raise dropped calls and handoff failures. Backhaul congestion can limit throughput even when radio capacity is sufficient. The operator should test before the event, monitor during it, and remove or redeploy the temporary equipment afterwards.

Issue	Control measure
High call blocking	Add temporary radio capacity and keep spare channel headroom
Co-channel interference	Coordinate the reuse pattern, power and antenna direction
Handoff failures	Prepare neighbour lists and test mobility paths
Backhaul congestion	Provision adequate temporary backhaul capacity
Temporary-site cost	Deploy only for event periods and redeploy equipment afterwards

OVERALL CONCLUSION

The assignment shows that cellular network planning is a balance between spectrum reuse, interference, traffic capacity, coverage, cost and mobility. Fixed cluster reuse gives a predictable baseline. Sectorisation improves spatial filtering by reducing unwanted signals. Dynamic channel assignment improves utilisation when traffic varies across neighbouring cells. Cell splitting and hierarchical deployment add capacity as the subscriber population grows.

For the given 351-channel system, N = 13 is selected as the planning candidate. It provides 27 channels per cell and about 19.265 Erlangs per cell at a 2% blocking target under Erlang-B. The ideal six-interferer SIR estimates are 12.11 dB, 16.08 dB and 24.04 dB for path-loss exponents 2.5, 3 and 4. The open-campus estimate sits only 0.11 dB above the 12 dB threshold, making field validation and sectorisation essential.

As the subscriber population grows, the network must add smaller cells and adopt a hierarchical macro–micro/small-cell layout. For 120,000 subscribers the first-order estimate needs about 187 cells at the selected per-cell traffic limit, so a macro-only network is insufficient for the full planning horizon. The recommended design combines fixed reuse, sectorisation, dynamic channel borrowing and selective cell splitting.

For the four-hour event, offered traffic is 1,200 Erlangs in 1.5 km², about 3.3 times the township-wide average density. The preferred response is temporary modification using microcells or cells-on-wheels, sectorised antennas and carefully controlled channel borrowing, which provides event capacity without the continuing cost of permanent densification.

The final network should be accepted only after measurements confirm the required SIR, outage, blocking and handoff performance under realistic user distributions. The mathematical models give a sound planning foundation, while field surveys and reproducible simulation are needed to validate the deployed network.

REFERENCES

- Mobile Computing – ITA03 Assignment-1 (CO1), supplied assignment brief and planning constraints.
- Mobile Computing – Assignment-1 reference solution supplied for formatting and analytical guidance.
- Standard cellular network planning concepts: hexagonal frequency reuse, Erlang-B traffic modelling, co-channel interference, sectorisation and dynamic channel assignment.
- Python 3 documentation for implementation of mathematical calculations and iterative procedures.